

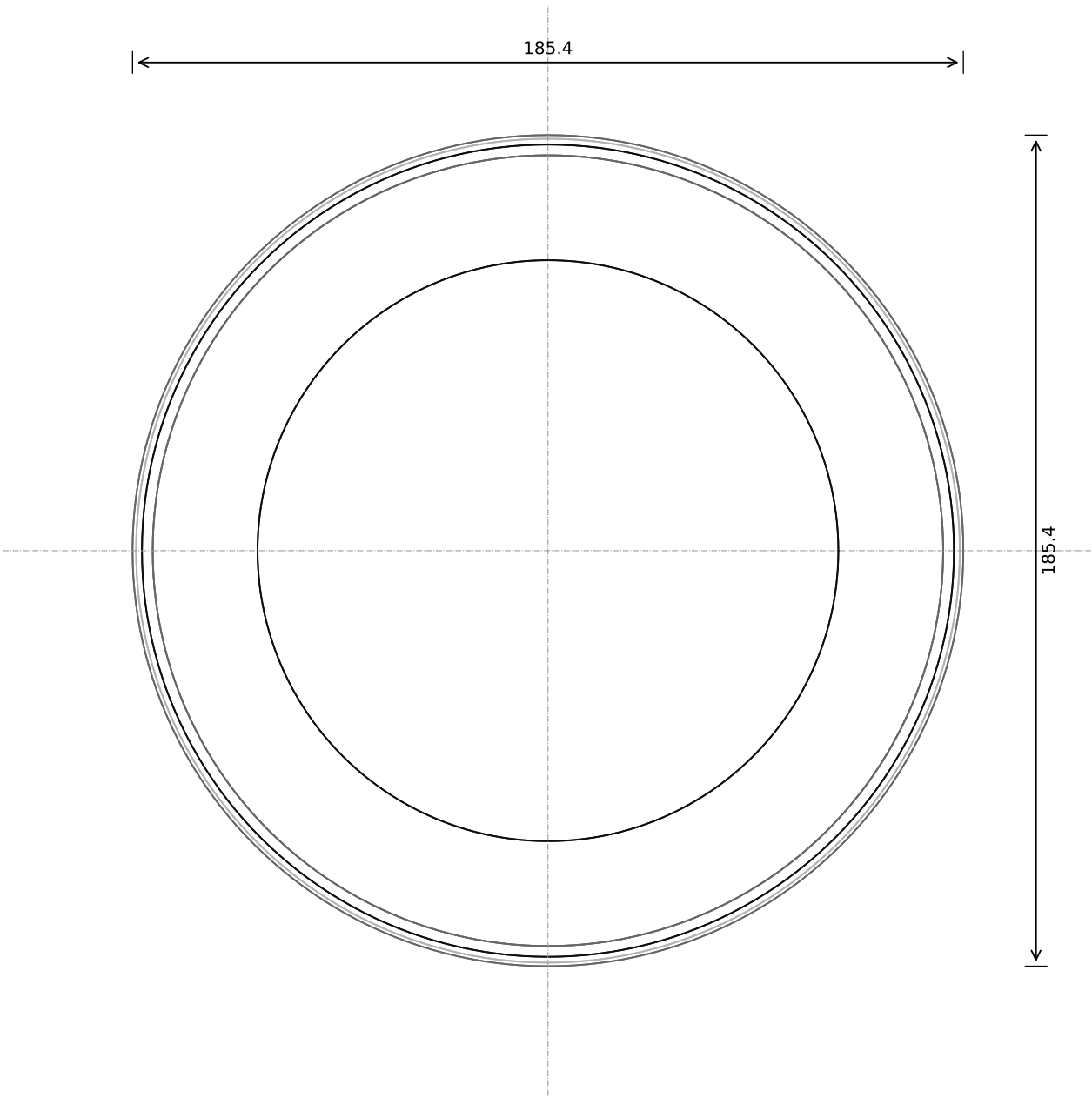
the 60 — v16 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-08. Stack: plate 8.0, adapter 14.78-16.38 (socket to 23.68), code ring under the crown at 31.53, seat 26.0, glass 26.50-28.48, lens to 30.98, knob top 34.38, overall 35.88 with the pad. Ø185.

1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board thickness and part heights: 1.6; parts 2.5, underside 1.0, flex connector 2.0	outline, holes, socket size and position PUBLISHED (datasheet drawing)
10. display connect board: 30.0 x 46.0 x 1.2, connectors 2.0	part of the adapter kit; size scaled from its photograph, hole positions assumed at the corners
11. Pi standoffs and washers: M2.5 x 12.0 hex, 0.5 washers	the adapter stack: closing-plate land / washer / Pi / standoff / adapter
12. audio / touch boards: envelopes	electrical layout not done
13. bought breakouts: TMC6300-BOB 25.4 x 20.3; DRV2605L 20.0 x 15.0; 74AHCT125 20.0 x 15.0; hole patterns	the vendors' pages where published; hole positions and part heights assumed
14. microcontroller: ESP32-S3-Zero class 25.0 x 18.0, USB-C 3.2 tall	RULING pending (question 33): a Pico 2 does not fit the tray
15. converter height: 7.6	Pololu D24V90F5: outline published, the inductor's height not
16. V-groove bearing: V623ZZ: V 1.2 deep, 0.4 root flat, 90.0 deg	the ridge is derived from these - measure Ryan's listing and re-run
17. block spring: Ø4.0 x 16.0 free, 0.8 N/mm	about 4 N seated; the bench sets it (the band slip limit is 120-160 mN.m at 4 N)
18. light-sensor breakout: sensor 3.0 from the edge	Adafruit 4162, from Ryan's photograph
19. panel jack body: Ø9.0 x 14.0	the listing's drawing to fetch
20. Mill-Max spring-pin block: 5.1 x 2.6 x 3.0, pins Ø1.5 x 2.2 compressed	867-22-002-70-501010; datasheet to fetch
21. diffuser gasket: 0.3	closed-cell foam ring under the lip
22. halo liner: 0.4 floor, 3.4 wall	white PETG; a white coating on the ring is the alternative (throw 4.0 instead of 3.6)
23. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
24. right-angle barrel plug: Ø8.5 body	measure the brick's plug
25. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
26. MOTOR_BAND_T: 0.6	carried from v9
27. servo pushrod: 3 mm bridge to the tab	carried from v9; servo on the plate in a frame
28. LED strip: 4.0 wide x 1.2 thick, face down	a 4 mm 2020 strip (HALO-OPTICS); the channel is 4.4 wide - a 5 mm strip needs the wall 1.0 further in
29. blower inlet and outlet: inlet Ø24.0 in the face, outlet 21.0 x 7.0 from 1.5 up	from the datasheet drawing (outline, holes and leads PUBLISHED)
30. blower gaskets: 0.3 closed-cell foam	under the inlet face and the hood
31. closing-plate gasket: 0.2 die-cut	in a groove of the recess floor, on every sealing wall
32. Pi 5 fan connector current rating: 0.13 A drawn (0.20 max)	the connector sits in the vendor STEP; its rating is not published
33. encoder shim family: [0.7, 0.8, 0.9, 1.0, 1.1]	printed; the height is chosen on the bench

Build notes in V16.md. The sheets that follow: one per made part.

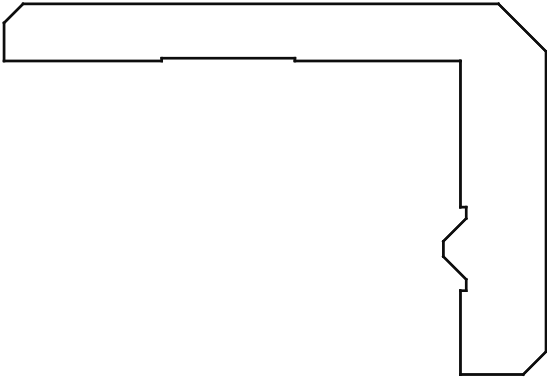
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 63 to 95, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- outside Ø185.4, bore Ø176.4 (0.45 to the panel's chin corners); wall 4.5
- top rim: lip inner edge Ø128.4, 1.0 inner chamfer, 25 flat, 2.5 outer chamfer (RULING 6 Sep: rim >= 20); lip 3.0 thick, top at z 34.38
- outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
- v16 V-RIDGE on the bore: relief band r 88.50 z 19.3-23.7; the ridge 90° flanks from its base (3.2 wide, z 19.9-23.1) to the crest r 87.30 (0.8 flat) - three V623ZZ V-groove bearings straddle it
- code RING recess 0.15 deep in the crown's underside, r 72.5-79.5 (v13: the encoder reads the crown, not the bore - the bore is plain apart from the V-groove)
- diamond knurl 82 starts x 2, 1.0 deep, helix 36.9°, 3 whole rows z 16.9-31.1; skirt bottom z 14.9
- wall under the knurl root 3.5

the 60 — v16

Knob

Cadrane · sheet 1 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)
Process: print top face down, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: knob_body
Overall 185.4 x 185.4 x 19.5
Volume 81.96 cm³
Scale: see each view (fit)

[illegible]

SECTION B-B (side, plane $x = 0$; y across, z up)

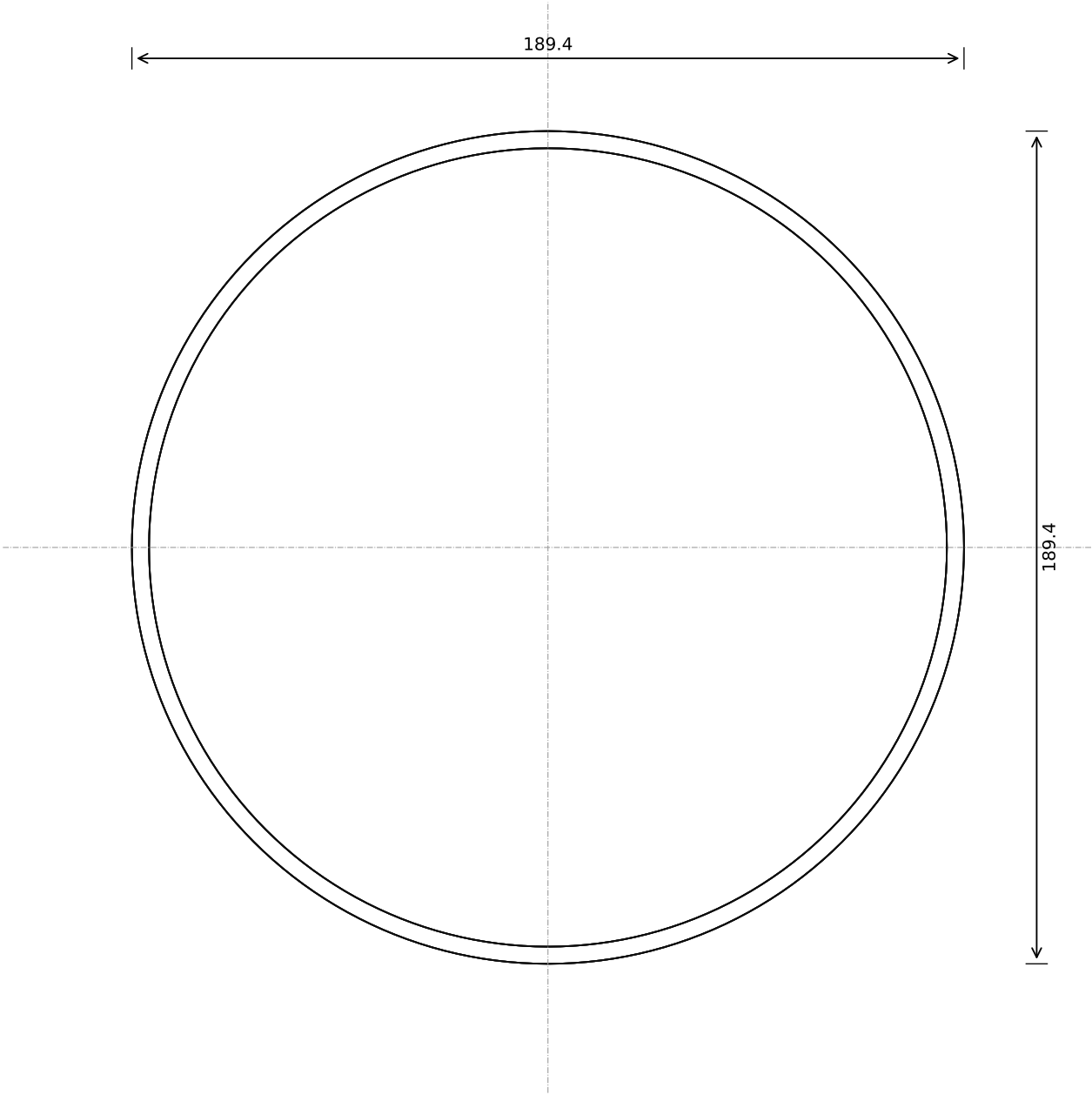


The side view shows a vertical shaft with a horizontal flange at the top and a horizontal base. The shaft has a central hole. The flange has a central hole and a smaller hole on the right side. The base has a central hole and a smaller hole on the right side.

1. wall in TWO radii: r 84.80-86.40 from the plate to z 13.8, a 50° ramp inward to r 82.80-84.40 at z 16.2, then up to the seat (the knob is fitted 2.4 off-centre and its far-side is 1.0 off-centre)
2. seat flange r 62.0-84.40, z 24.0-26.0: its top face is FLAT with 0.3 chamfers on both edges; three Ø2.0 H7 through-holes (ream) at [70, 210, 330]°, r 67.25 for the display's cable
3. HALO (HALO-OPTICS): the strip's ledge r 84.9-90.50, z 13.20-14.3 (the 4.0 mm strip stuck face DOWN under it, r 86.6-90.6, 55 LEDs, 2.64 A full white, firing at the white line)
4. two FIXED wheel posts Ø8.0 at r 82.3, az [30, 150], to z 19.0, Ø2.9 press holes for the Ø3 m6 x 10 pins; the SPRUNG wheel's channel at 270°: walls 2.0 at t ±4.2-6.2 from the plate
5. pillars Ø7 at r 80.80, az [160, 240, 328], to z 16.0; M3 inserts from below (the plate screws) - nothing screws in from above
6. encoder at 335.0°, r 76: two radial slots through the flange at t ±4.75 over M2 inserts in Ø5.0 bosses under it; LRA pad at 258.0° (r 81.6 to the wall) with the contact carrier
7. flex slot 32 x 2.5 through the seat at 0°, offset -4; 102 vent slits 1.0 x 4.5 at z 18.5-23.0, 2.5° grid; motor relief at 90° (the bell's window 1.0 clear, the collar's 2.0, open up to 1.0)

bowel pins, each in a Ø5.0 boss 4.0 under the flange er 3.6 below): the lip r 90.50-92.70, z 13.5-14.3 clamps the opal ring through a 0.3 Material: PETG (prototype) / PA12 (production) 58.0 to the wall, z 8.0-14.0, roof z 13.2-14.0 with a 3.2 slot for the pin (travel 4.0 in / 0.5 out), end wall r 58.0-60.0 with a Ø2.4 hole at z 10.5 for the relea Process: v16: print INVERTED - the seat flange's top face on the bed, 0.20 1's min. beside its flange tail at 252.5° (two ribs from r 74.5 to r 0.4) grooves each, 45° to the flange); halo tail notch at 325.0°, the far-end feed's at 190.0° static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	gasket, 0.3 radial clearance Part: internal structure Overall Ø5.4 and Ø5.4 x 16.0 Volume 97.48 cm³ Scale: see each view (fit)
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PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



5.7

DETAIL C (section A-A, x 65 to 97, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- r 90.8-94.7 (3.9 wall), z 7.5-13.2 (5.7 tall); vertical outer face standing 2.0 proud of the knob and plate (Ø189.4)
- 0.3 chamfer on all four edges
- stands in the rim ring's 0.5 rebate (r 90.50 outward, floor z 7.5) on the 0.8 shelf; 0.3 radial clearance to the rebate's wall and the ledge; clamped axially by the lip through a

the 60 — v16

Halo ring (opal acrylic)

Cadrane · sheet 3 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

0.3 foam gasket - never hard between two machined faces

Material: cast opal acrylic, 6 mm sheet (Perspex Opal 030 ~68 % / PLEXIGLASS® 030 ~83 % class), CNC routed, annealed 80 °C 2

Process: machined, not printed (HALO-OPTICS \$6)

Overall 189.4 x 189.4 x 5.7

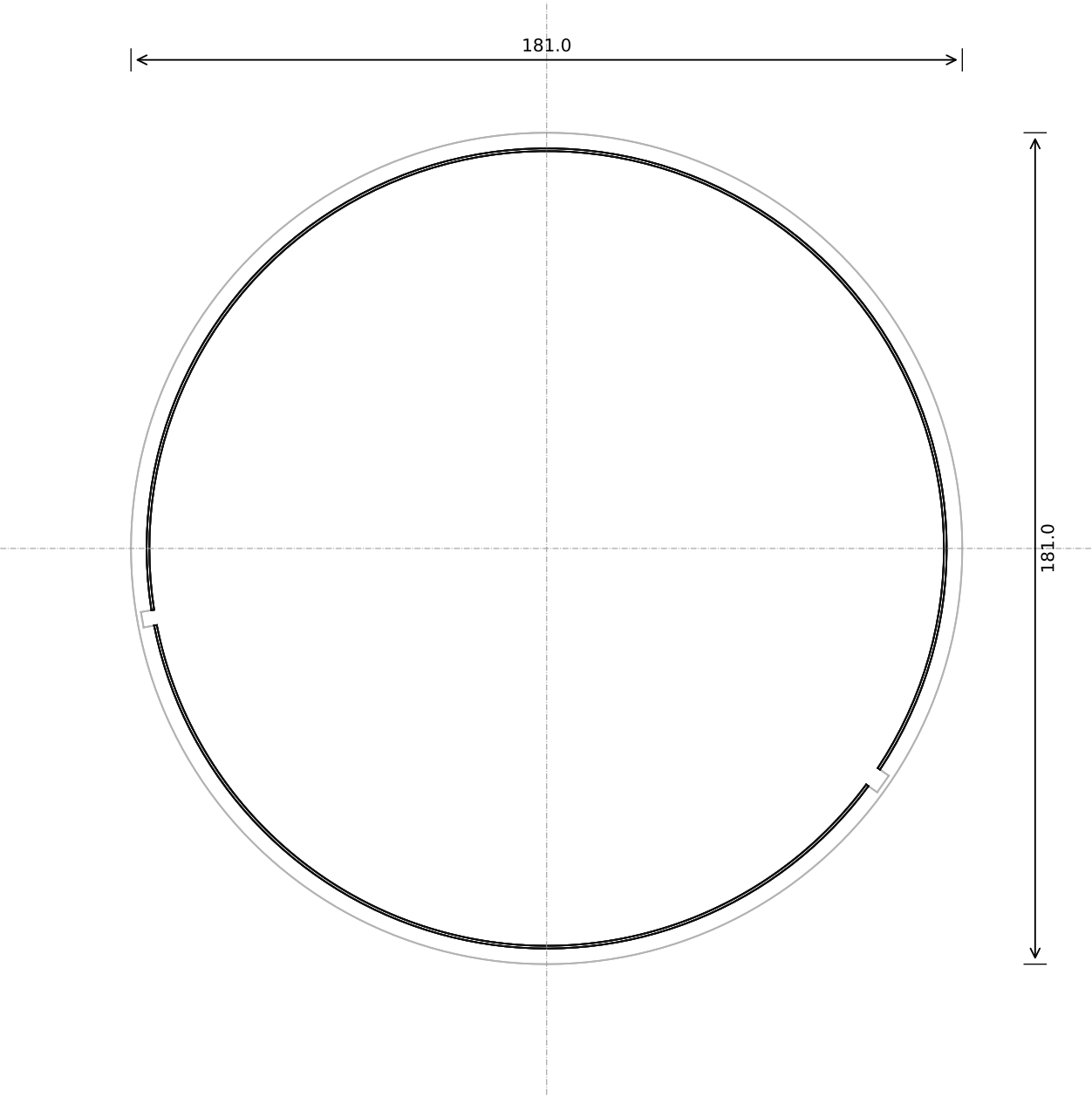
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

Volume 12.85 cm³

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



3.40

DETAIL C (section A-A, x 61 to 93, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. L-section ring: floor r 86.5-90.50 x 0.4 on the rim ring's top, wall r 86.5-87.1 up to z 11.4 against the structure's wall (0.6 under the strip); cut at the two tail notches
- 2. HALO-OPTICS §9: the floor and walls must be white and diffuse - bare stainless is specular

the 60 — v16

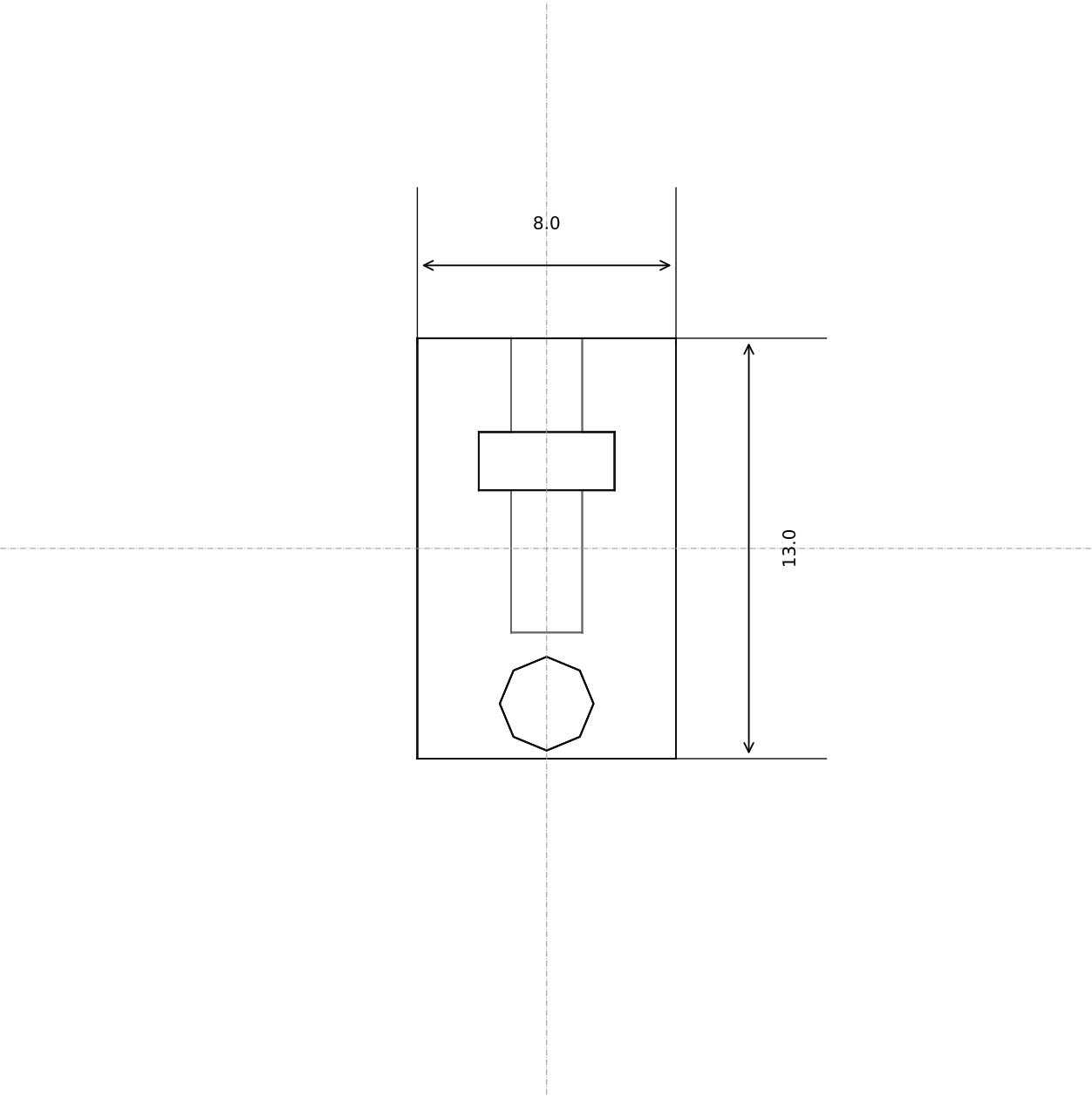
Halo liner (white)

Cadrane · sheet 4 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

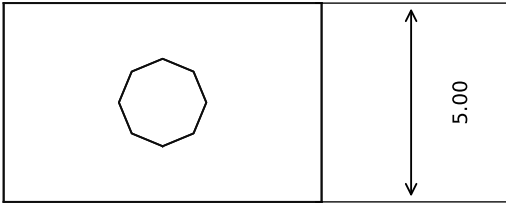
Material: white PETG
Process: v16: print flat on its floor, 0.16 mm; no supports
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: halo_liner
Overall 181.0 x 181.0 x 3.4
Volume 1.85 cm³
Scale: see each view (fit)

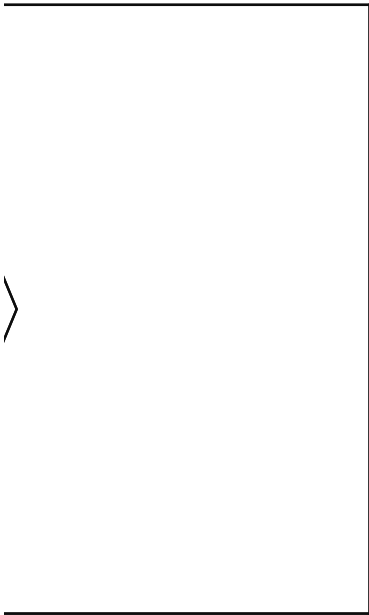
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



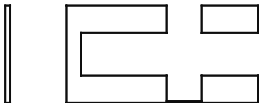
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 2 to 6, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 13.0 (radial) x 8.0 x 5.0 on the plate top at 270°, r 71.0-84.0 nominal; Ø2.9 press hole for the Ø3 m6 x 16 axle pin 1.7 from its outer end (the pin to z 24.2)
- 2. Ø2.2 hole from the inner face at z 10.5 for the M2 x 18 release screw; an M2 nut in a 4.2 slot from the top, 3.0-4.6 behind the face
- 3. spring Ø4.0 x 16.0 free, 0.8 N/mm on the screw between the channel's end wall and the block: 4.0 N seated, 7.2 N parked
- 4. 0.2 to the channel's walls and roof; slides on the anodised plate

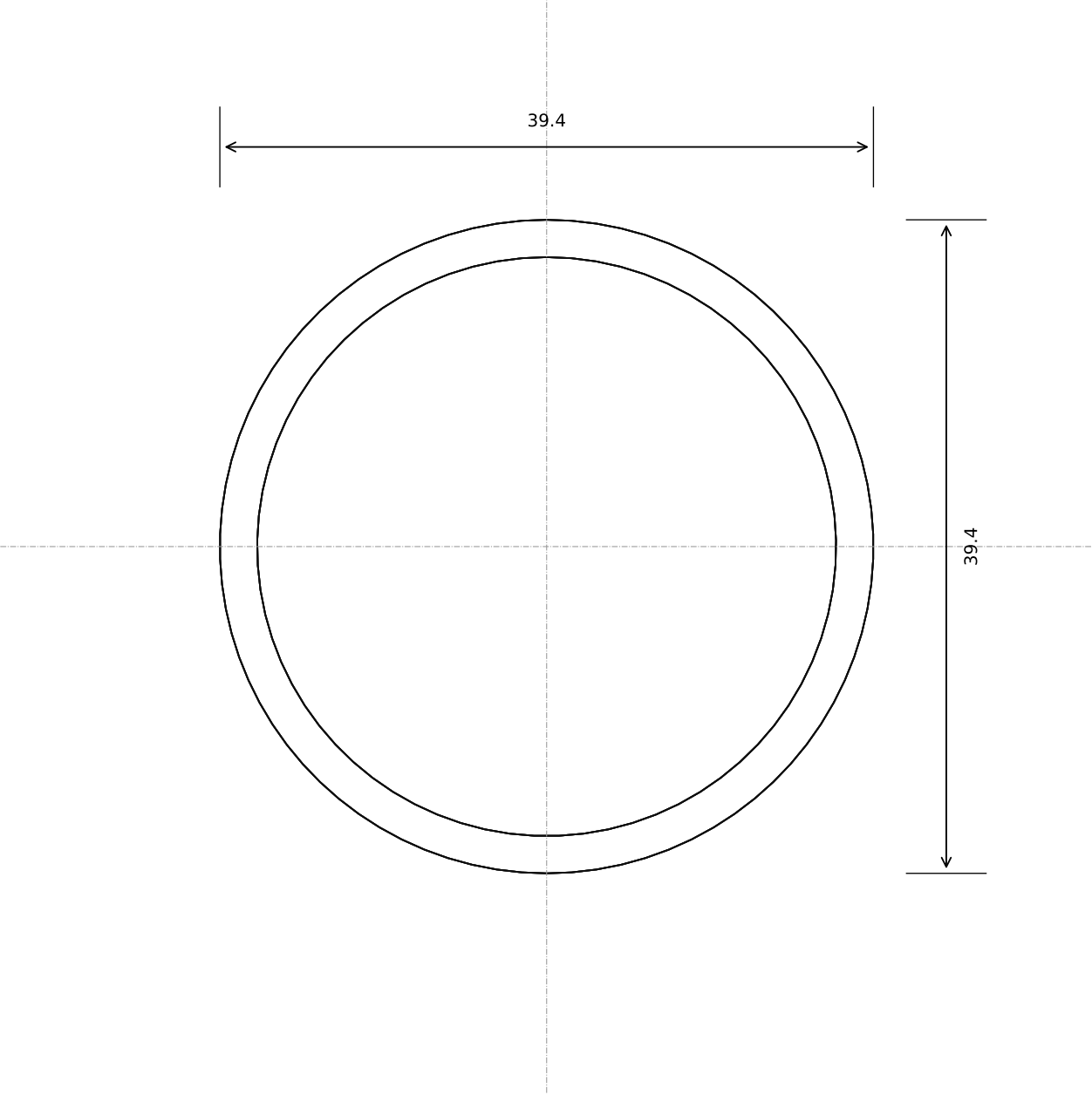
the 60 — v16Sprung wheel block

Cadrane · sheet 5 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG, 4 perimeters
Process: v16: print flat on its bottom face, 0.12 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: wheel_block
Overall 8.0 x 13.0 x 5.0
Volume 0.42 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 10 to 22, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. ID 35.0 -0.05 (press) x OD 39.4 x 3.5, on the bell's top (z 14.7-18.2); 0.3 lead-in chamfer
- 2. the 0.6 silicone band on it at z 14.90-17.8 reaches the bore (r 88.20); the bell itself stops at r 86.0, inside the wall's outer face and 0.5 inside the halo liner
- 3. bore/collar ratio 4.48

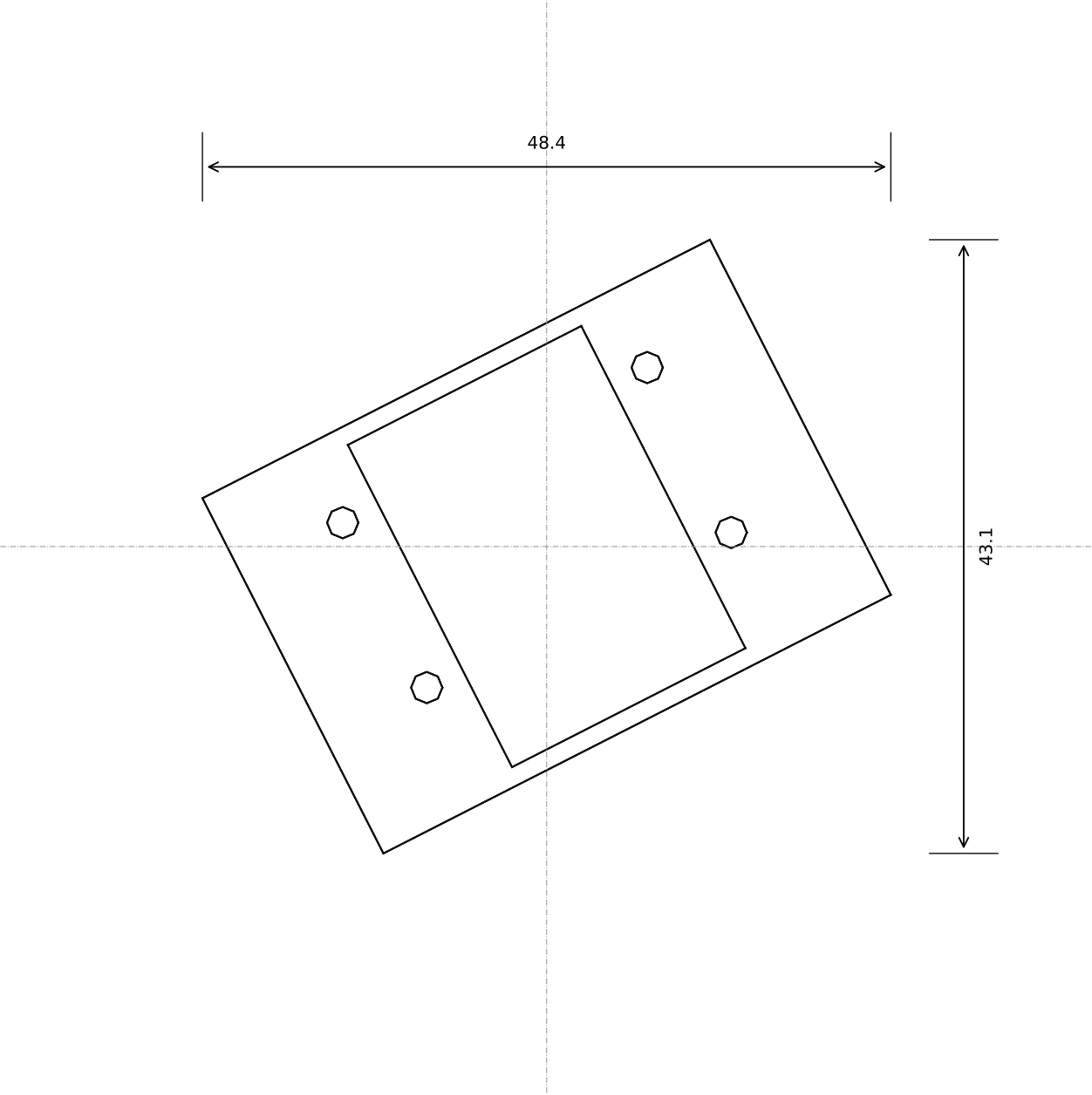
the 60 — v16 Drive collar (on the bell)

Cadrane · sheet 6 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

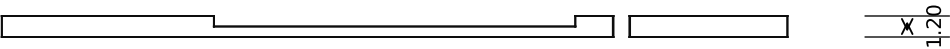
Material: PETG (prototype) / POM (production)
Process: v16: print axis vertical, 0.12 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: drive_collar
Overall 39.4 x 39.4 x 3.5
Volume 0.91 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 26, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 40.0 x 28.0 x 1.2 at z 16.6-17.8, centred 0.0 toward the ports of the audio board's centre; four Ø2.2 on the audio's stud pattern (24 x 13); on 4.4 tube spacers above the audio's nuts
- a 0.6 pocket: the ESP32-S3-Zero class MCU (25.0 x 18.0) between the four stud nuts, USB-C toward the west end (the level shifter is on the plate at az 322.0, r 69.5)
- no screw touches the MCU: the keeper's rails hold it by its edges

the 60 — v16

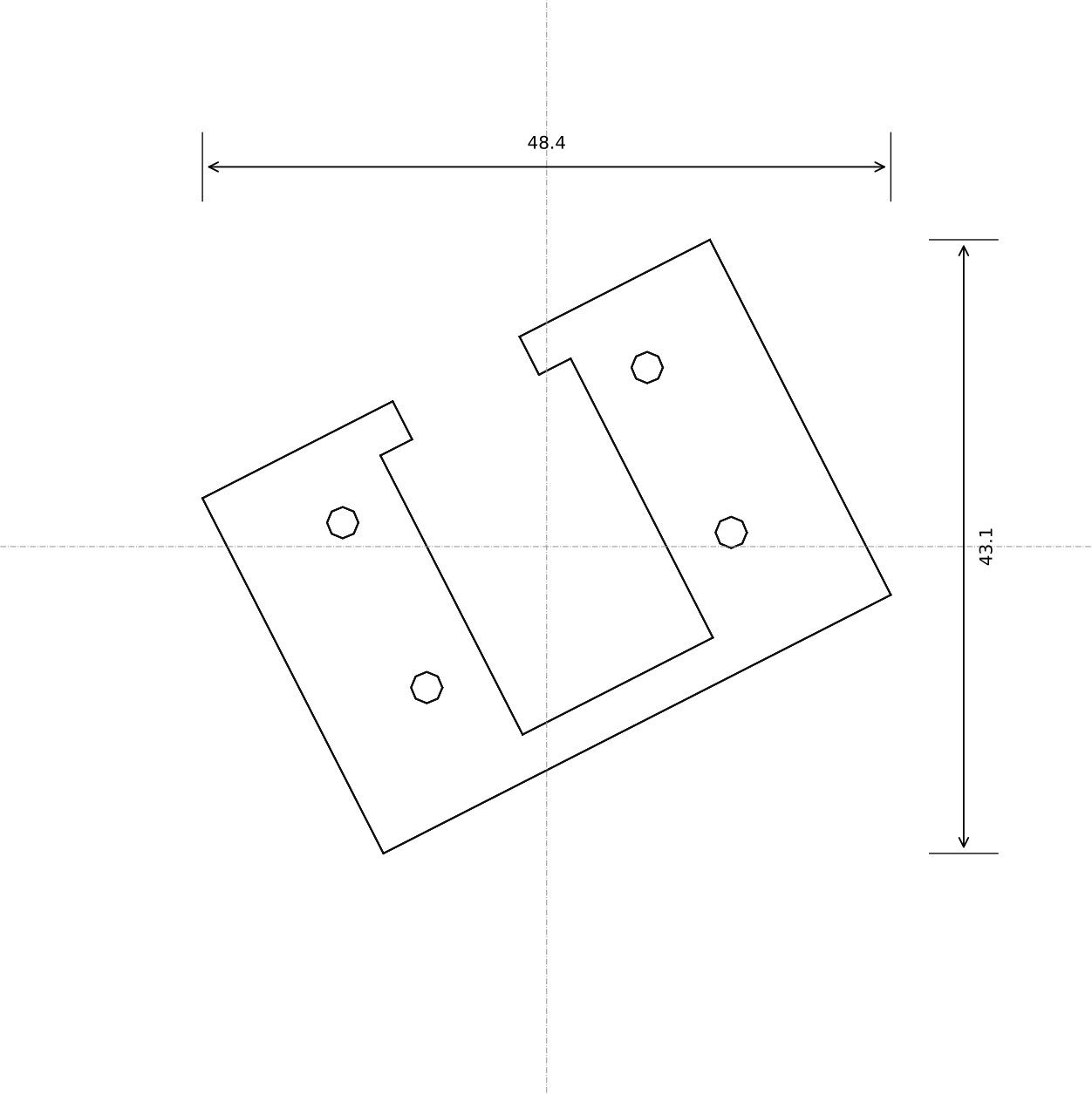
MCU tray (over the audio board)

Cadrane · sheet 7 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v16: print flat on its underside, 0.16 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: mcu_tray
Overall 48.4 x 43.1 x 1.2
Volume 1.05 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 26, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. 40.0 x 28.0 x 1.2 on the tray (z 17.8-19.0), clamped by the audio studs' top nuts; a window over the MCU's parts (1.5 rails); its USB-C end open

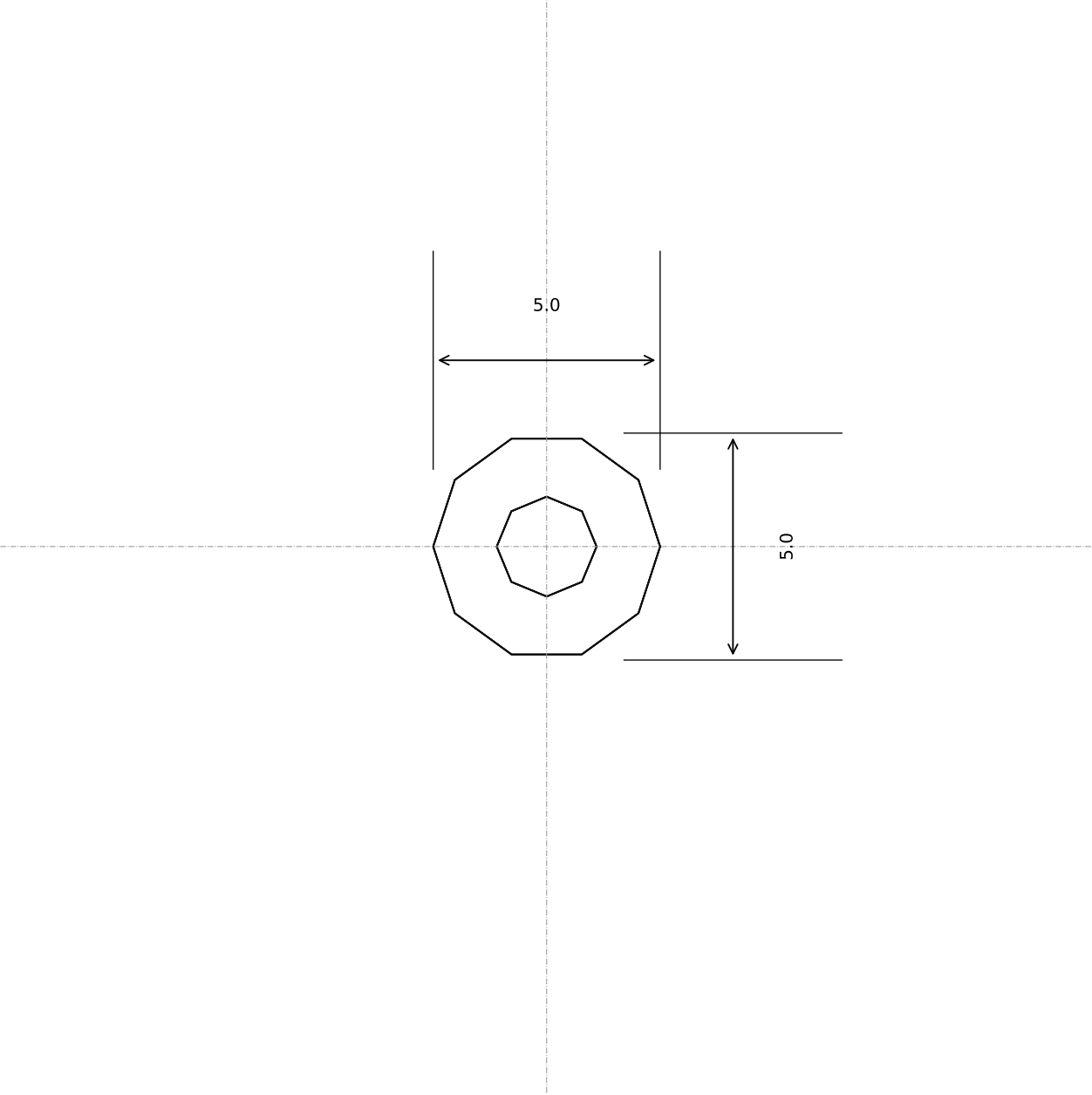
the 60 — v16 Tray keeper

Cadrane · sheet 8 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

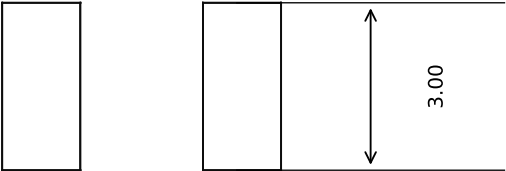
Material: PETG
Process: v16: print flat, 0.16 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: tray_keeper
Overall 48.4 x 43.1 x 1.2
Volume 0.89 cm³
Scale: see each view (fit)

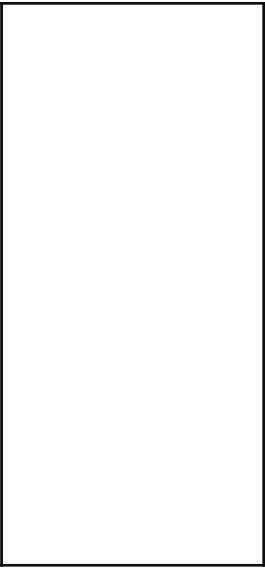
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



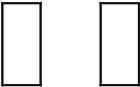
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 1 to 4, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. Ø5.0 tube, bore 2.2: 3.0 under the driver and haptic breakouts, 1.0 under the audio board, 4.4 between the audio's nuts and the tray
- 2. the countersunk M2 stud passes through; the nut above the board clamps the stack

the 60 — v16

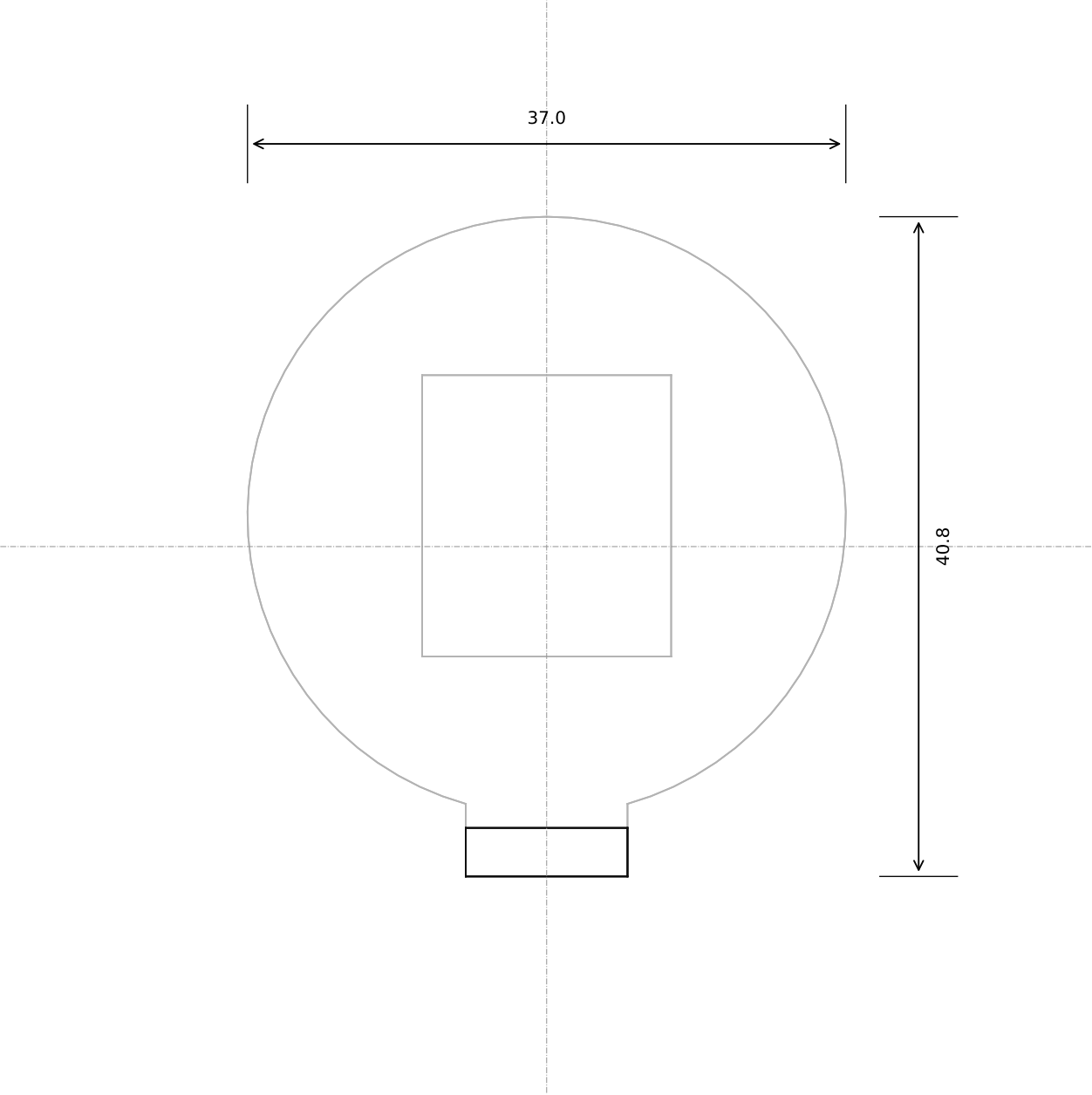
Board spacer (family)

Cadrane · sheet 9 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

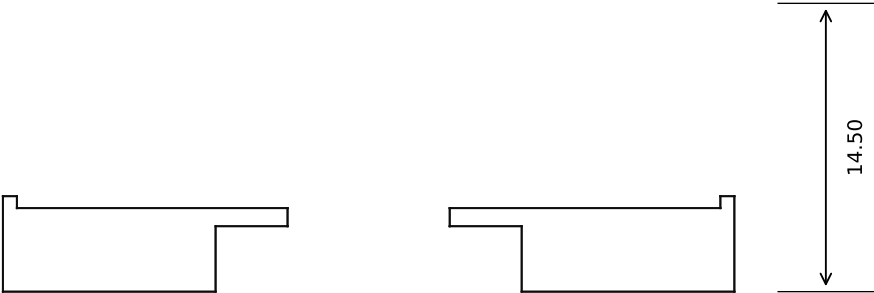
Material: PETG
Process: v16: print axis vertical, 0.12 mm; 3.0 x 4, 1.0 x 4, 4.4 x 4 mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: board_spacer
Overall 5.0 x 5.0 x 3.0
Volume 0.05 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

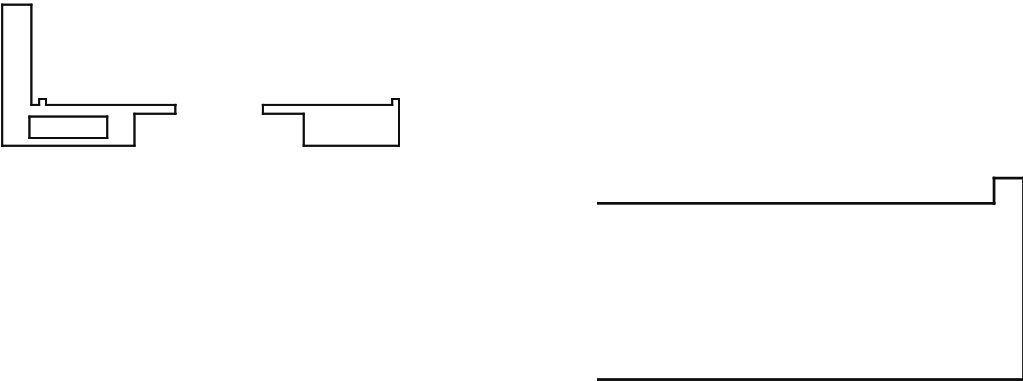


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. shoe $\varnothing 37 \times 4.2$; locating rim $\varnothing 35.6/37 \times 0.6$
- 2. pocket $17.4 \times 15.4 \times 3.3$ for the MT6701 module (v16); sight hole $\varnothing 9$; a 2.4×2.2 notch through the shoe's inboard wall for the sensor lead
- 3. push tab 3.0×10.0 to $z\ 14.5$ (v11: 28.5 through the deck; the servo is now beside it on the plate)
- 4. travel 2.4 radial in the plate's stadium hole

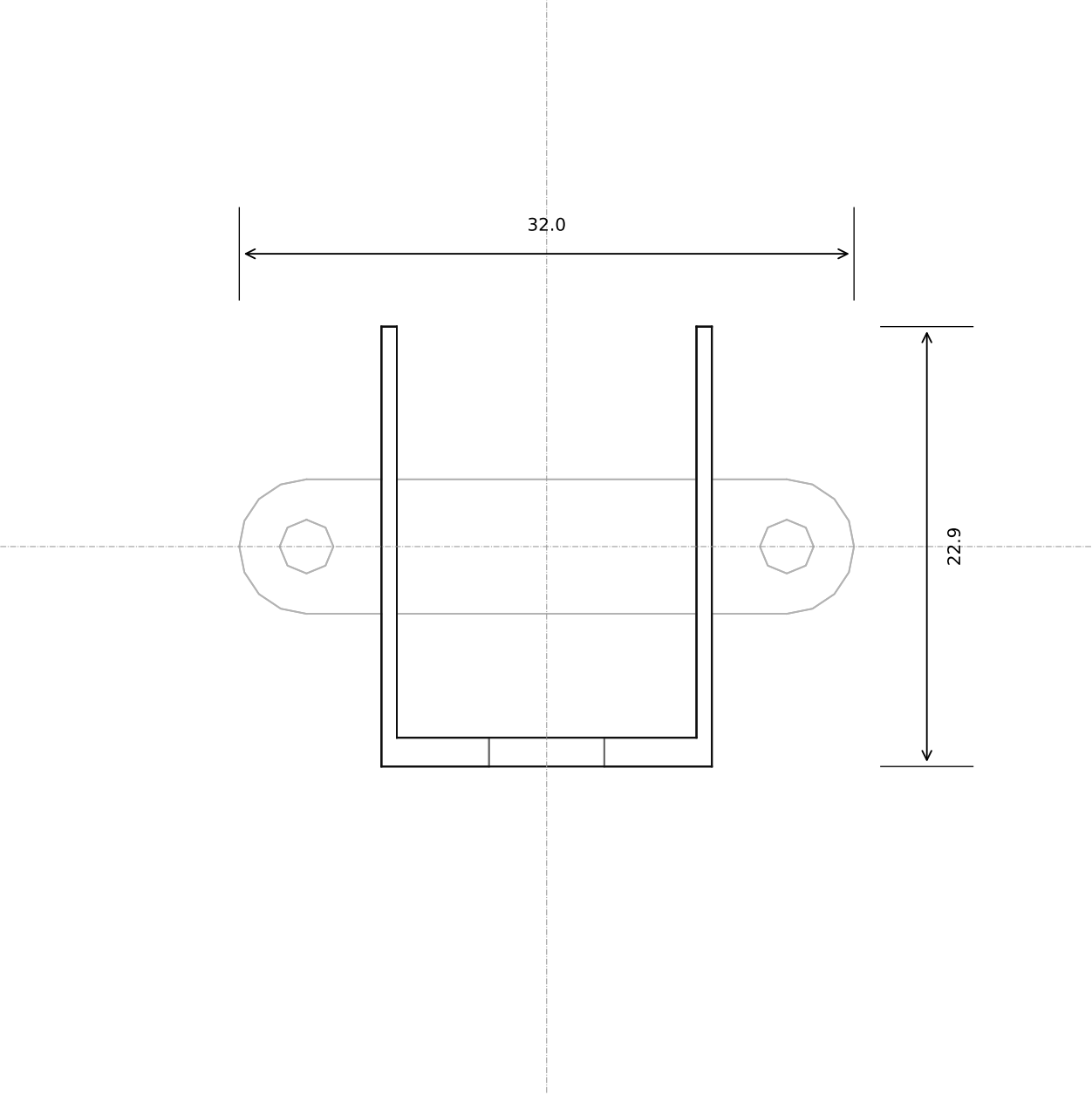
the 60 — v16

Motor carriage

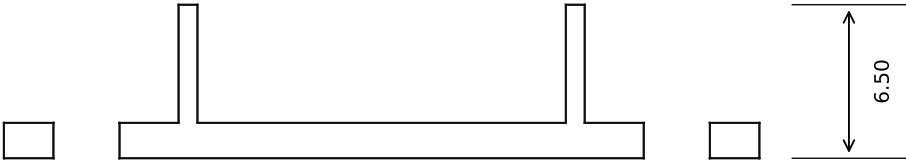
Cadrane · sheet 10 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG	Part: carriage
Process: print standing on the shoe, 0.16 mm; no supports (12 mm bridge over the 37 mm $\varnothing$ hole)	Overall 37 mm $\varnothing$ × 40.8 mm × 14.5 perimeters for the tab
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 4.06 cm ³
	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

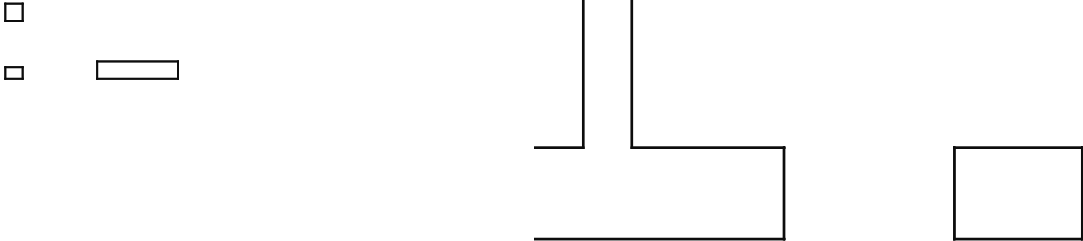


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 18, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. floorless frame round the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, 6.5 tall; the servo lies on the plate at r 21.8-43.2, az 90
- 2. open toward the carriage tab; lead notch in the inner end wall
- 3. 2 ears at t ±12.5, M2.5 x 4 into tapped blind holes in the plate's web

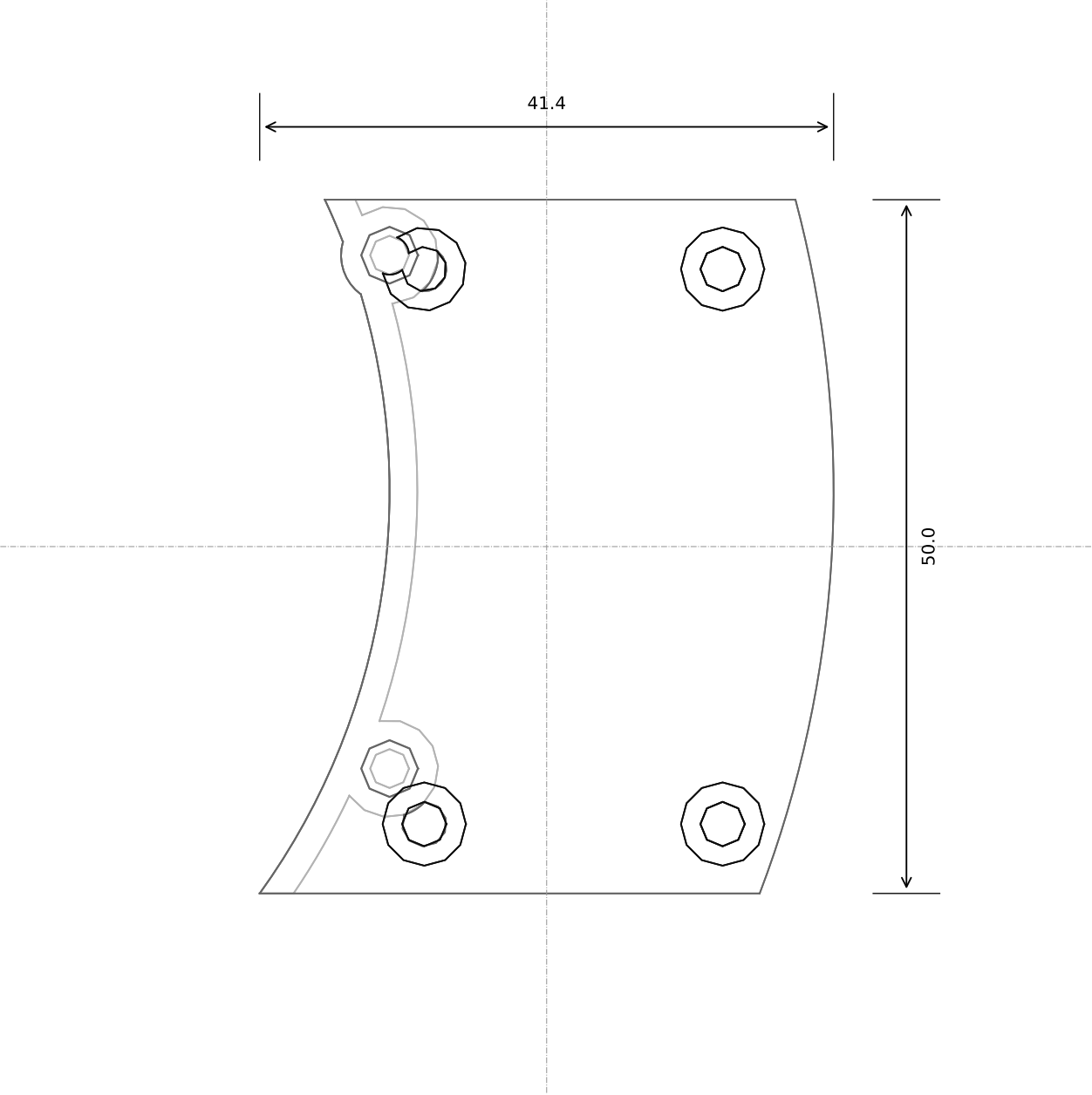
the 60 — v16 Servo frame (on the plate)

Cadrane · sheet 11 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

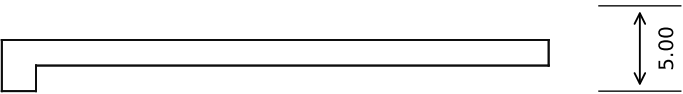
Material: PETG
Process: print flat, 0.20 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: servo_mount
Overall 32.0 x 22.9 x 6.5
Volume 0.64 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 10 to 23, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. slab 1.5 thick at z 9.5-11.0, r 50-82, t -29..21, bridging the port slot; its outer end rests on the port face's rail (z 9.5)
- 2. foot strip r 50-52 on the plate; two csk ears at [(49.6, 17.0), (49.6, -20.0)], M2.5 into the web
- 3. four Ø6 bosses 2.0 tall with M2 inserts for the connect board (30.0 x 46.0, holes 3 in from its inner corners, 5.5 from its outer)
- 4. the -y inner corner cut back to x 45.5 (the audio board's USB plug is there)

the 60 — v16

Connect-board bracket

Cadrane · sheet 12 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

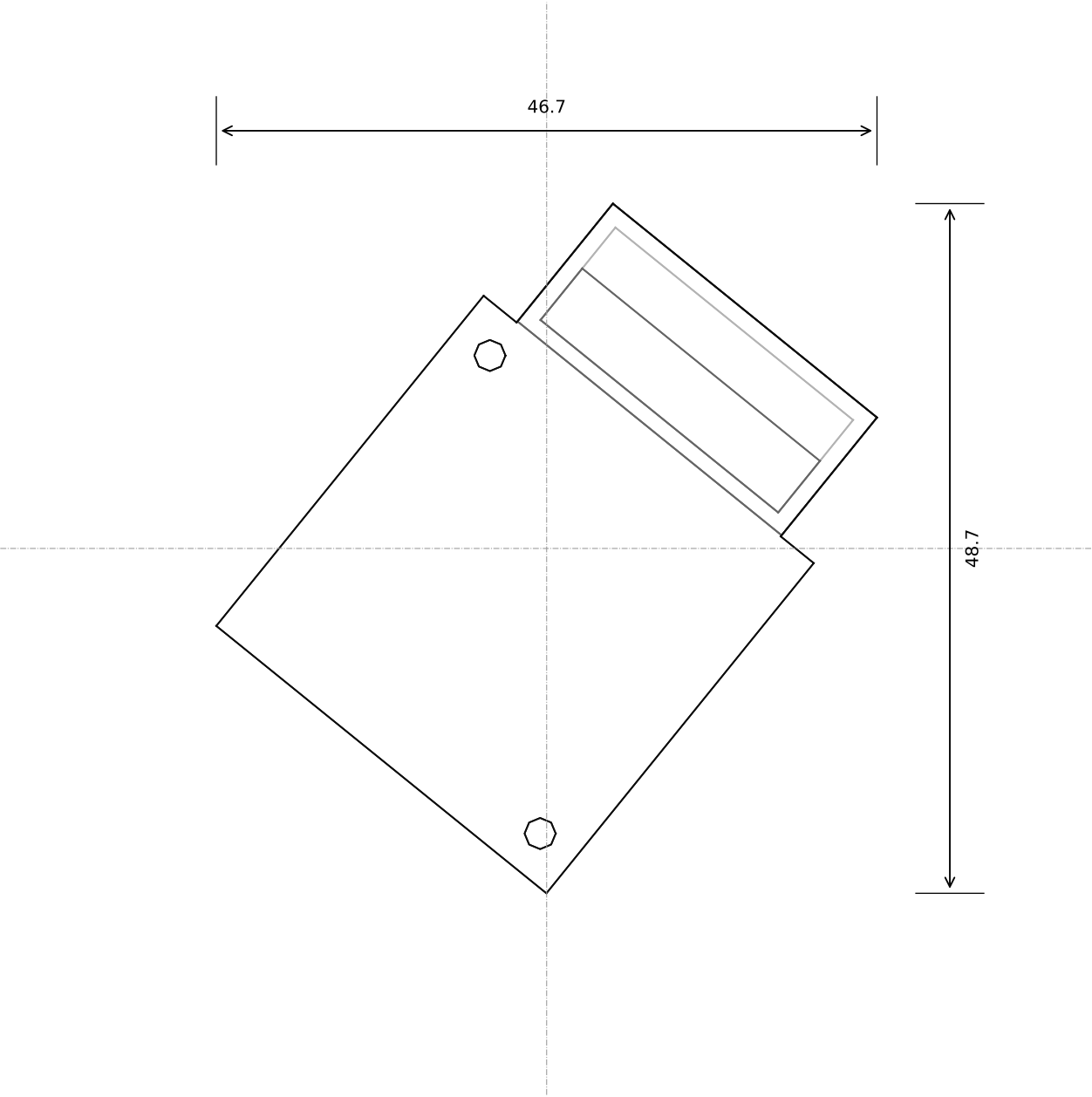
Material: PETG

Process: print standing on its straight -y edge (slab vertical, feet and bosses clear by 41.40 x 20.00 x 5.00 supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: connect_bracket

Size: 41.40 x 20.00 x 5.00
Volume 2.80 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 12 to 25, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. plate 30 x 30 x 1.2 on the blower's top (z 18.3-19.5) - v16: a plain clamp plate, no board on it; two Ø2.2 through-holes on the blower's 24 x 24 diagonal: M2 x 16 pan heads through the plate and the blower into the plate's piers
- 2. hood neck: 24 wide, r 66-76.8, walls 1.2, open at the bottom over the trench and toward the blower's outlet (21 x 7); on a 0.3 foam gasket on the plate top
- 3. blower at az 51, r 51 (Delta BFB0305HA-C 30 x 30 x 10, inlet down over the Ø22 hole in the web)

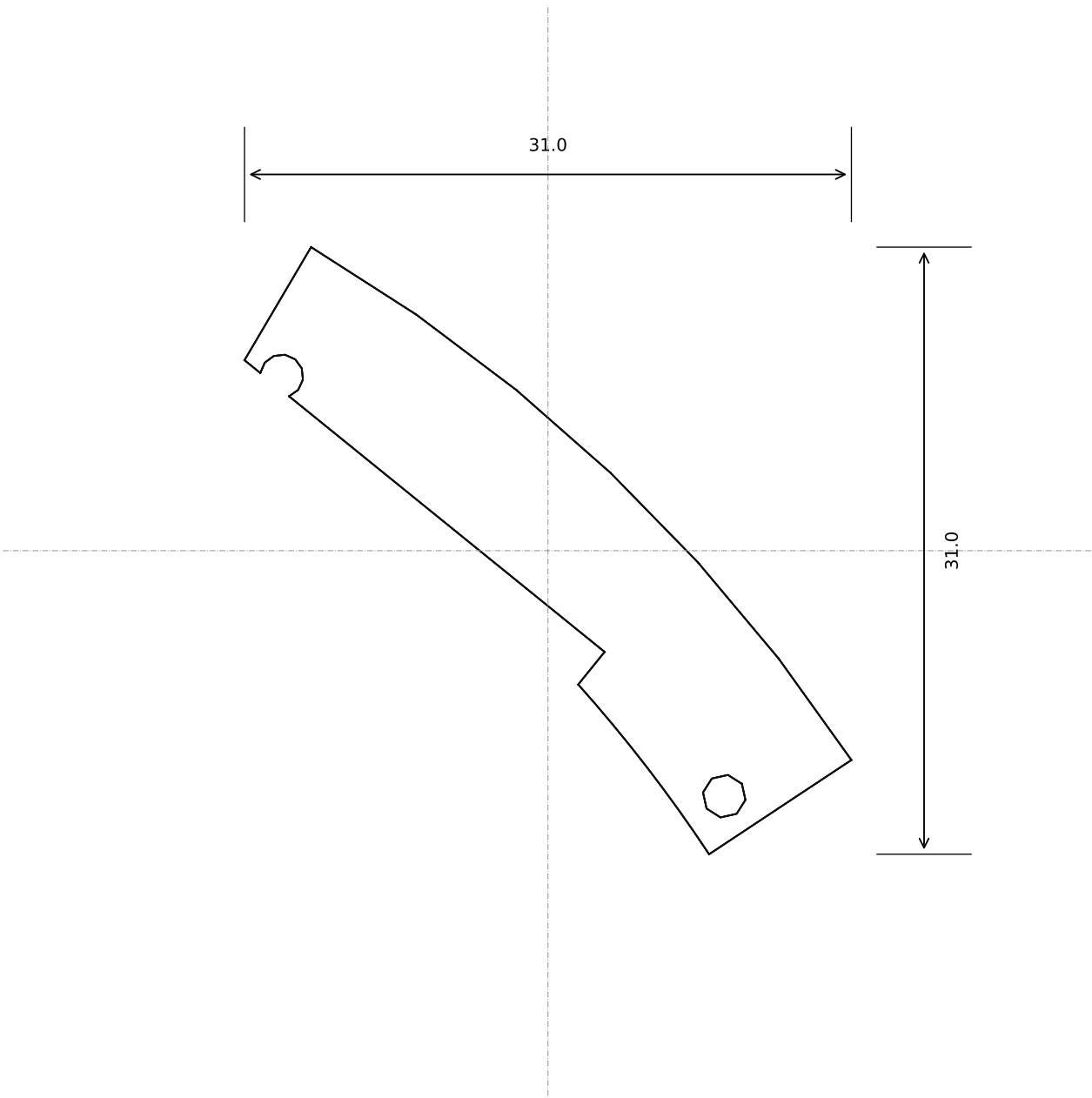
the 60 — v16

Blower saddle + hood neck

Material: PETG
Process: print on its flat top face, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: **blower_saddle**
Overall 46.7 x 48.7 x 11.2
Volume 2.76 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 17, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):
1. 2.5 thick at z 8.3-10.8, az 33.5-59.5, r 75.8-84.5; butts the saddle's neck (0.15 clear); two Ø2.2 at az [35.0, 58.0], r 78.1: M2 x 5 into the ring's top; on the same 0.3 gasket

the 60 — v16

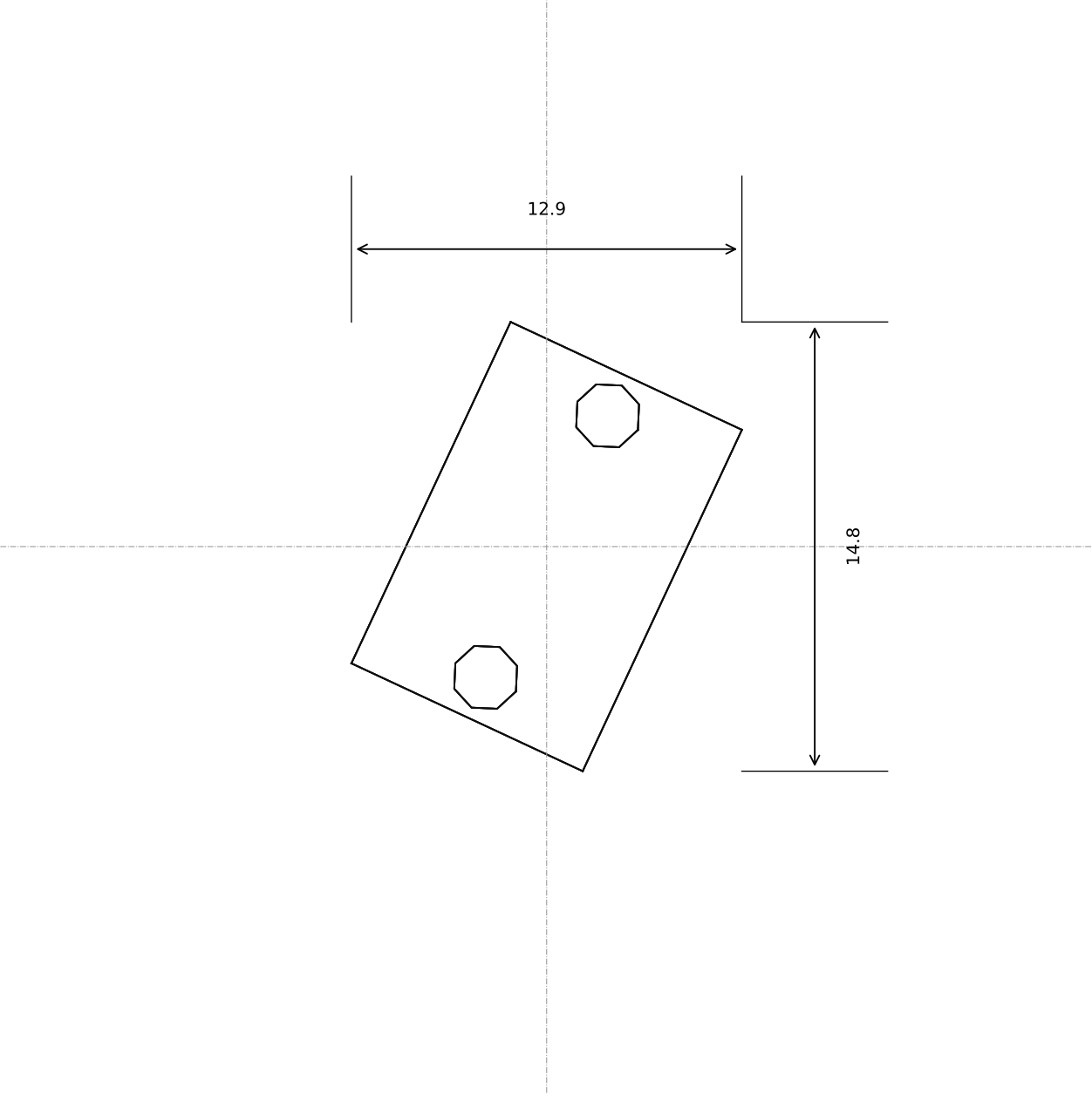
Hood lid (over the trench foot)

Cadrane · sheet 14 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

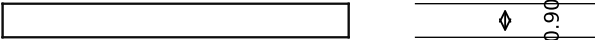
Material: PETG
Process: v15: print flat, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: hood_lid
Overall 31.0 x 31.0 x 2.5
Volume 0.68 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 8, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):
1. 8.4 x 12.4 x 0.9 (nominal; family [0.7, 0.8, 0.9, 1.0, 1.1]), two Ø2.2 at ±4.75; sits on the seat flange under the encoder board, sets the 2.0 optical gap

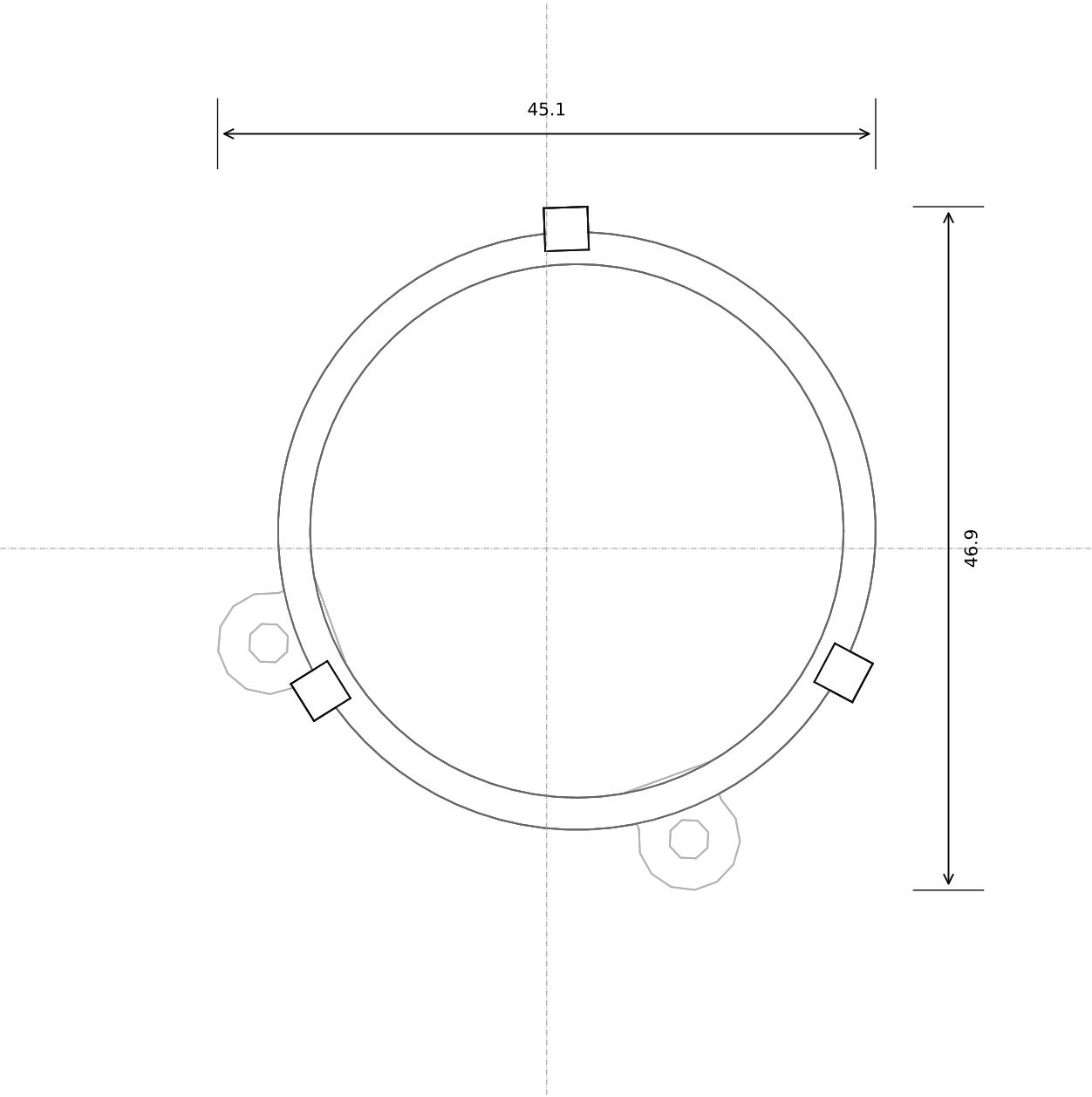
the 60 — v16 **Encoder shim (family)**

Cadrane · sheet 15 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v15: print flat, 0.10 mm; one of each height in ENC_SHIM_FAMILY
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: encoder_shim
Overall 12.9 x 14.8 x 0.9
Volume 0.09 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 25, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- ring r 18.3-20.5 x 4.45 under the Ø40 flange
- three snap fingers at [92.0, 212.0, 332.0]° from the speaker's centre (device azimuths)
- two ears at r 22.5, [200.0, 290.0]° from the centre; M2.5 x 4 from above into the plate's web (v11: M3 from below)
- speaker centre (-36, 47) - az 127°, r 59.2

the 60 — v16

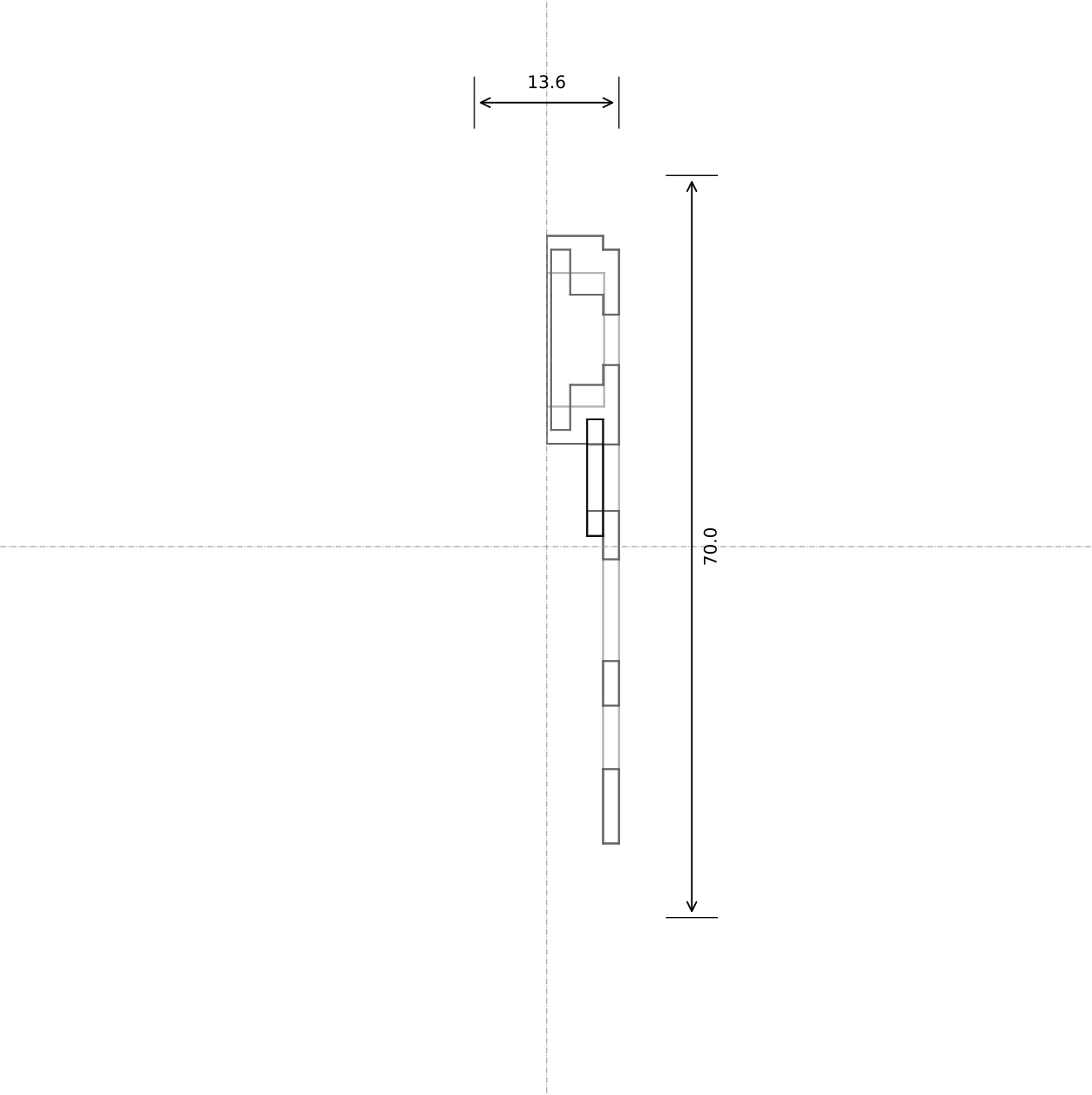
Speaker cradle

Cadrane · sheet 16 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

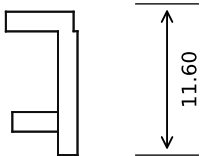
Material: PETG
Process: print flat, 0.20 mm; no supports (fingers full height)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: speaker_cradle
Overall 45.1 x 46.9 x 7.2
Volume 1.44 cm³
Scale: see each view (fit)

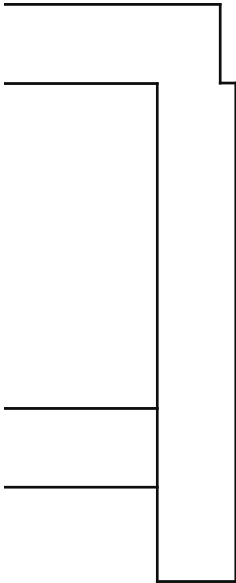
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 56.0 wide x 1.5 thick at r 83.30, z -1.5-8.0
- 2. barrel nose Ø6.2 at t -18.0, z 3.5; USB-C opening at t -6.0, z 4.5; jack Ø6.3 at t 6.5, z 4.6; light aperture Ø4.8 at t 19.5
- 3. shelf for the USB-C board; a 3.0 slot holder for the Adafruit 4162 light-sensor breakout (16.6 x 16.5, standing from z 1.0 to 17.5, sensor at the hole) with a pocket for its sensor package; a 3.0 boss round the panel jack's Ø6.3 hole; rail along the top on the plate (1.5 tall, 70 wide, inner edge r 79.3, flat front in the faces plane, Material: PETG

the 60 — v16

Rear port face

Cadrane · sheet 17 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Process: print lying on its outer face, 0.16 mm; no supports (the socket holder is printed on the other side of the plate)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

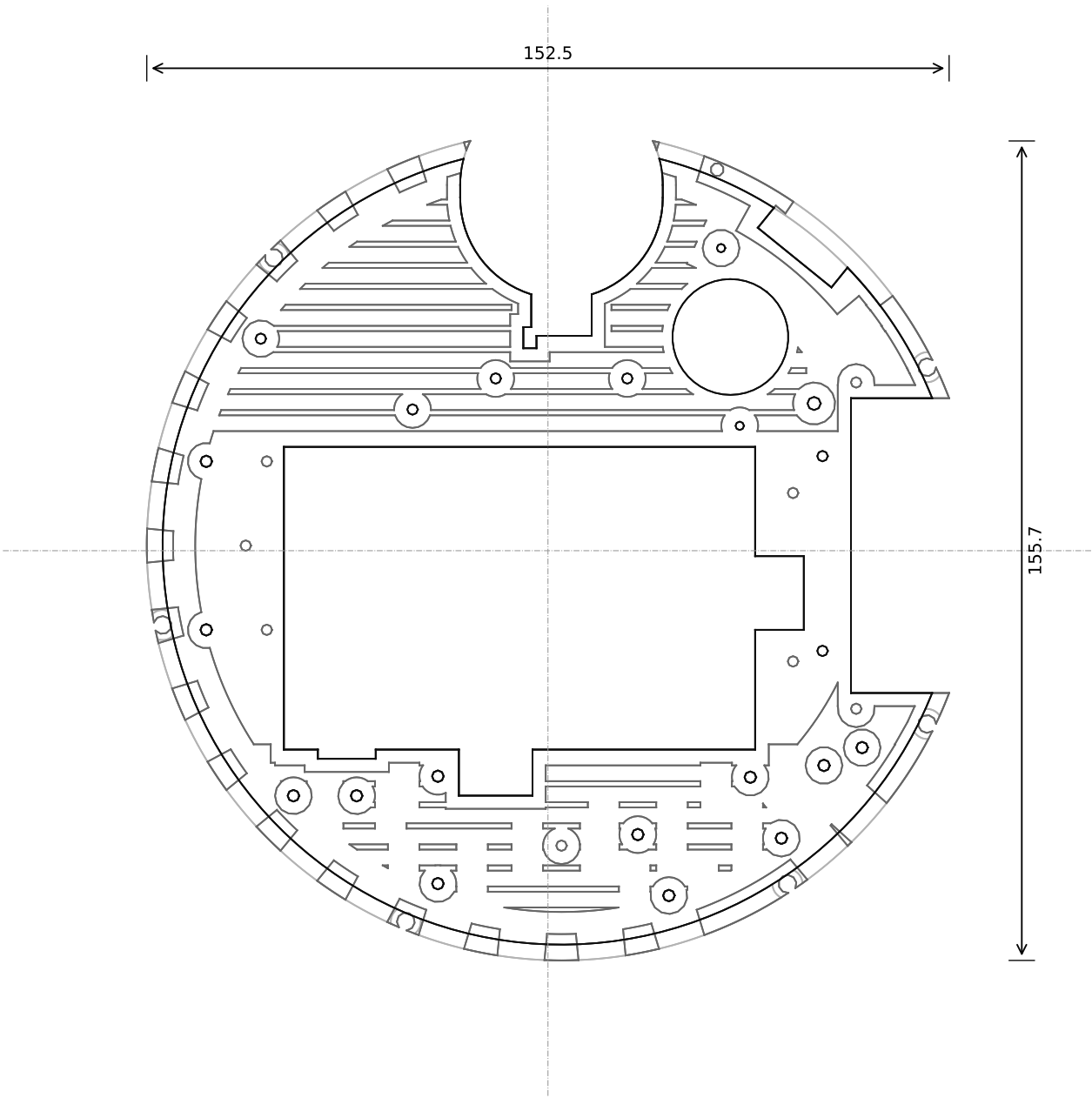
Part: port_face

Overall Bbox: 70.0 x 11.6

Volume 1.65 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 46 to 78, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. $r < 78.8 \times 8.0$: 2 web (z 6-8), 5 duct (z 1-6) closed by a 1 mm plate in a recess on a 0.2 gasket in a groove; $\varnothing 7$ piers 2.5 into the duct under every screw from the top; rebate r 75.8-78.8, z 5-8 for the ring's flange on a 5 shoulder, 7 x M3 csk from below at r 77.30; the ring's bond screw M3 tapped 3.0 into the shoulder at az 67.0
2. duct: collector r 70-73.8 (front and sides), 21 straight channels 3 x 5 on a 4 pitch beside the Pi window, rear plenums r 59+; 2.5 walls round every through-cut; intake passages as closed tunnels z 1-4.5 through the duct wall and shoulder (18 x 9 wide) + 2 exhaust tunnels 10 wide at az [310.0, 318.0]
3. BLOWER (v15, fitted): $\varnothing 22$ inlet hole through the web at az 51, r 51; discharge trench 18 wide from r 70.4 out through the duct wall and the shoulder, widening to az 37-56 (open-topped, the hood seals it); the 2 mm wall at r 68.4-70.4 keeps it off the suction side
4. Pi window (Pi at (-52, -38), with the USB-C receptacle, HDMI plug and USB-A plug notches - v16: no plug-head notch, the Pi is powered through its header), carriage hole, port slot 56.0 wide; 6 x M2.5 tapped blind 3.5 deep from the top (servo frame, connect bracket, speaker cradle) + 2 x M2 (the blower) + THE chassis bond screw
5. v16: flat top face - no ribs (the two v15 ribs are deleted)

the 60 — v16

Base plate - aluminium core (the heatsink)

Material: 6082 aluminium, CNC two-sided; EXTERNAL faces black hard anodized	Part: base plate
Process: machined, not printed (print a PETG stand-in without the duct for to test the fit)	Overall 152.5 x 155.7 x 8.0
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 45.60 cm ³
	Scale: see each view (fit)

Technical drawing of a semi-circular architectural element, likely a section of a dome or a large arch. The drawing shows a cross-section with a semi-circular profile. The outer edge is defined by a series of small, rectangular blocks, possibly representing masonry or a decorative frieze. The inner edge is a smooth curve. The drawing includes two dimensions: a horizontal dimension of 185.2 and a vertical dimension of 189.4. The drawing is oriented with the semi-circular arc on the left and the straight edge on the right.

8.00

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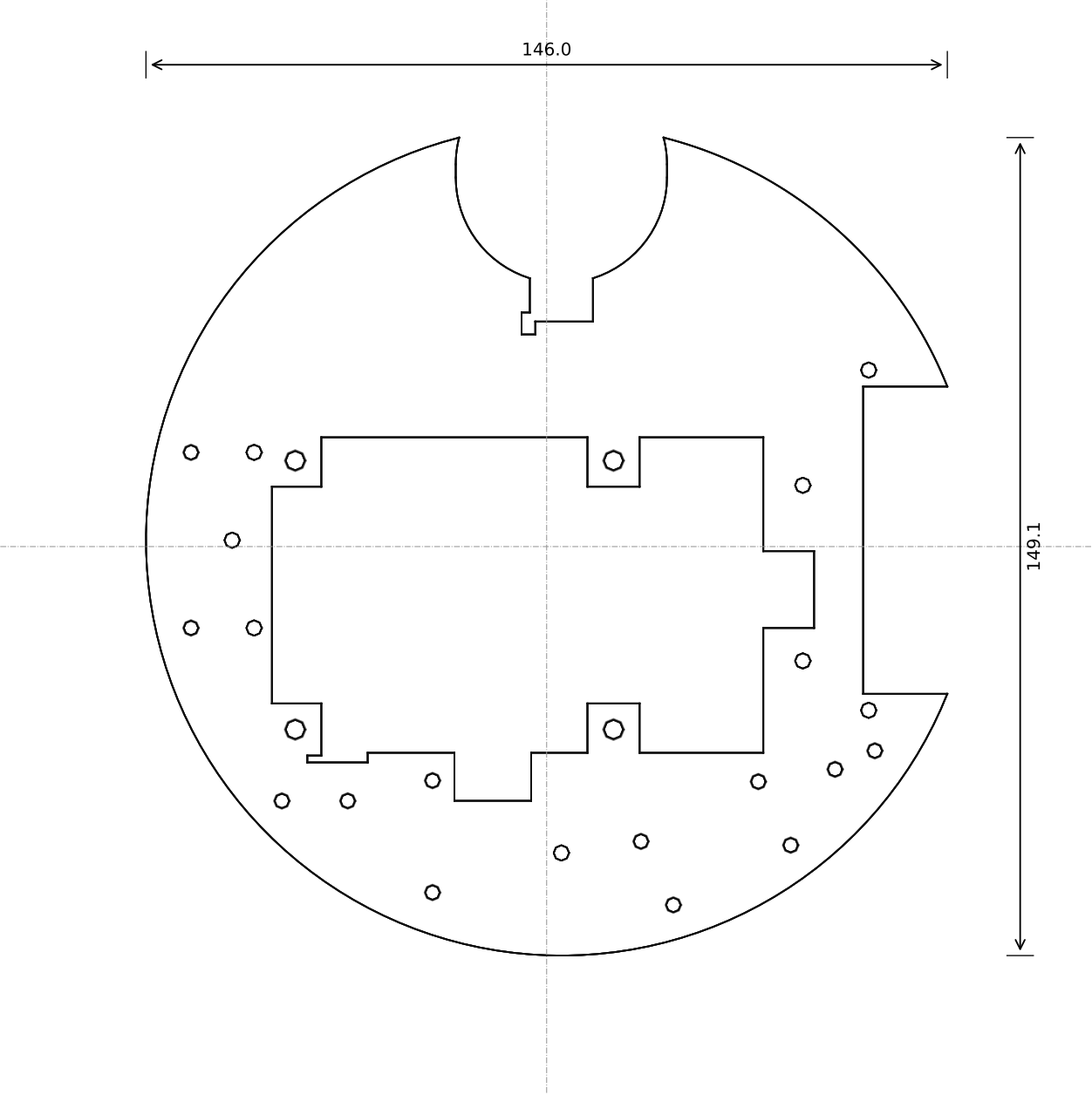
D

1. r 78.8-92.7 x 8.0; 90 obround openings 2.0 x 4.5 (z 1.55-6.05) on 3° through the 3.7 outer wall, az 1.5 + 3k, mirrored above 0-180, none across the port face (342-18): 64 intake (60-300) + 26 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both of its edges; 0.35 above the openings; chamfered mouth	Cadrane - sheet 19 of 20 - 2026-09-08 - drawn from src/params.py + model.py (build123d) the outer wall (r 89.00-92.7) opens into it all round both arcs; the blower's trench: flange notched and inner land cut away az 37-56 to r 83.00, the groove at az 67.5, r 77.30 (M3 from the top into the core, star washer under the head, both faces bare) Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 chamfers and 0.35 above the openings) the -y circuit; the +y circuit exhausts through the trench Process: machined, not printed Overall 185.2 x 189.4 x 8.0 Volume 32.11 cm³
2. underside groove behind the openings: r 84.50-89.00 (intake arc) / r 82.50-89.00 (exhaust arcs), z 0-6.55, walls 1.2° at 60 and 300; the pad closes it; the 0.8 undercut under the	
3. inward flange r 75.8-78.8, z 5-8, 7 x M3 tapped blind from below at r 77.30, az [26.0, 78.75, 135, 191.25, 247.5, 303.75, 334.0]; the dedicated BOND screw's Ø3.4 clearance	
4. passages across the inner land (r 78.8 to the groove), 3.5 deep, the pad closes them: 18 intake 9 wide (567 mm2) on the 11.25° grid, 2 exhaust 10 wide at az [310.0, 318.0]	
5. 3 x M3 csk from below at r 80.8, az [160, 240, 328] (the structure's pillars); 2 x Ø1.6 tapped M2 at r 81.8, t ±32 (the port face's rail); port slot 56.0 wide	
6. v16 (HALO-OPTICS): the opal ring's seat - a 0.5 rebate in the top from r 90.50 outward and a 0.8 shelf r 92.7-94.7 at z 6.7-7.5 standing 2.0 proud (0.2 polished chamfers on its edges)	

Rim ring (stainless)

Intake (60-300) + 26 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both o Cadrane · sheet 19 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build 123d) er the outer wall (r 89.00-92.7) opens into it all round both arcs; the blower's trench: (blade notched and inner land cut away az 37-56 to r 83.00, the groo le at az 67.5, r 77.30 (M3 from the top into the core, star washer under the head, both faces bare) Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 ch Blower's trench, 5° pitch mouth and both edges), then brush the (the -y circuit; the +y circuit exhausts through the trench) Process: machined, not printed its edges) 0.35 above the openings; chamfered mouths mm; tolerance ±0.1 unless stated; clearances: runnings 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Overall 185.2 x 189.4 x 8.0 Volume 32.11 cm³ Scale: see each view (fit)
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PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 43 to 75, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. disc r 75.7 x 1.0, flush in the core's recess on a 0.2 die-cut gasket (two pieces); same Pi window, carriage hole and port slot as the core
- 2. 12 x Ø2.2 countersunk from below (DIN 7991 M2 heads, 0.2 proud) for the boards' studs
- 3. four 9.5 square lands reaching into the Pi window at the Pi's holes [(-48.5, -34.5), (9.5, -34.5), (-48.5, 14.5), (9.5, 14.5)], Ø2.7 csk from below: the Pi and adapter stack screws to them (M2.5 x 6 into the standoffs)
- 4. 9 x M2.5 countersunk; unscrew to clean the fin channels

the 60 — v16

Closing plate (aluminium)

Cadrane · sheet 20 of 20 · 2026-09-08 · drawn from src/params.py + model.py (build123d)

Material: 1 mm aluminium, laser-cut, black anodised (outer face)

Process: laser-cut, not printed

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: closing_plate_AL

Overall 146.0 x 149.1 x 1.0

Volume 10.68 cm³

Scale: see each view (fit)